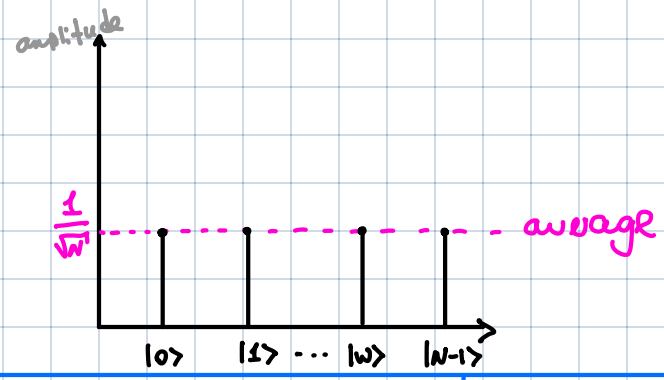
Amplitude Amplification

↳ the general idea behind the Grover's algorithm.

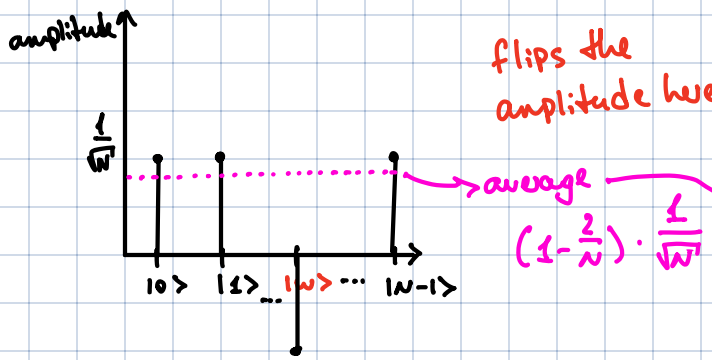
Let us have a look at the amplitudes at each step in Grover's algorithm.

Step 1. $|s\rangle : H^{\otimes n}|0\rangle^{\otimes n}$



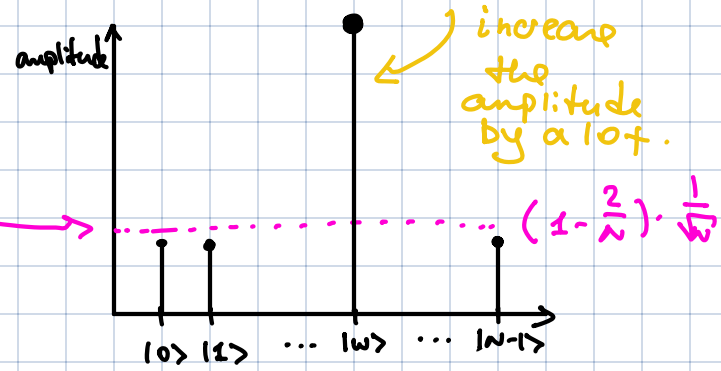
Step 2. Apply the oracle.

$U_f|s\rangle = (I - 2|w\rangle\langle w|)|s\rangle$.



Step 3. Apply the Diffuser V.

$V \cdot U_f \cdot |s\rangle = (2|s\rangle\langle s| \cdot I) \cdot U_f \cdot |s\rangle$



as for $|\psi\rangle = \sum_i \alpha_i |i\rangle$

↳ $V|\psi\rangle$ yields:

$(2|s\rangle\langle s| - I)\cdot|\psi\rangle$ ⊜

⊜ $\underbrace{2\cdot\frac{1}{N}}_{?}\cdot\sum_j |j\rangle\cdot\sum_k\langle k|\cdot\sum_i \alpha_i|i\rangle - \sum_i \alpha_i|i\rangle$ ⊜

⊜ $2\cdot\underbrace{\sum_k \alpha_k\cdot\frac{1}{N}}_{\text{average } \langle\alpha\rangle}\cdot\sum_j |j\rangle - \sum_j \alpha_j|j\rangle$ ⊜

⊜ $\sum_j |j\rangle\,\underbrace{(2\langle\alpha\rangle - \alpha_j)}_{\text{reflection of } \alpha_j \text{ about average } \langle\alpha\rangle}|j\rangle$

$\alpha_j = \langle\alpha\rangle + \Delta_j$

↳ $2\langle\alpha\rangle - \alpha_j =$ becomes

$= \langle\alpha\rangle - \Delta_j$

④

⇒ repeating steps 2 & 3 the amplitude of $|w\rangle$ will increase even further.

⇒ amplitude amplification.

What about when we have multiple marked elements?

Say we have M marked elements w_i, we define the winning state as

$$|w\rangle := \frac{1}{\sqrt{M}} \sum_{i=1}^{M} |w_i\rangle \rightarrow |w^\perp\rangle = \frac{1}{\sqrt{N-M}} \sum_{x \notin \{w_1,\dots,w_M\}} |x\rangle$$

$$\Rightarrow |s\rangle = \frac{\sqrt{N-M}}{\sqrt{N}} |w^\perp\rangle + \sqrt{\frac{M}{N}} |w\rangle = \cos\frac{\theta}{2} |w^\perp\rangle + \sin\frac{\theta}{2} |w\rangle$$

$$\hookrightarrow \sin\frac{\theta}{2} = \sqrt{\frac{M}{N}} \Rightarrow \theta = 2 \arcsin\sqrt{\frac{M}{N}}$$

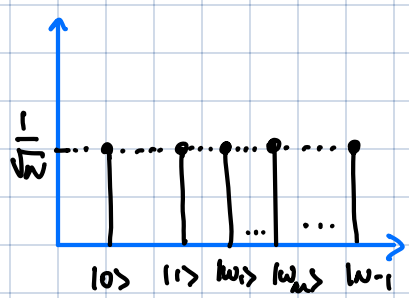
angle θ becomes larger!

$$\Rightarrow r = \frac{\pi}{4 \cdot \arcsin\left(\sqrt{\frac{M}{N}}\right)} - \frac{1}{2} = O\left(\sqrt{\frac{N}{M}}\right)$$

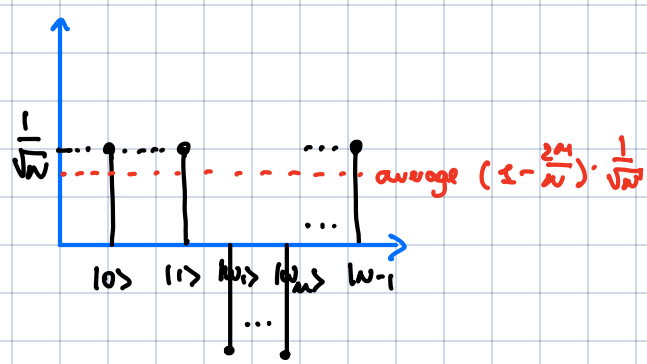
even faster.

we can see this speedup also when looking at the amplitudes:

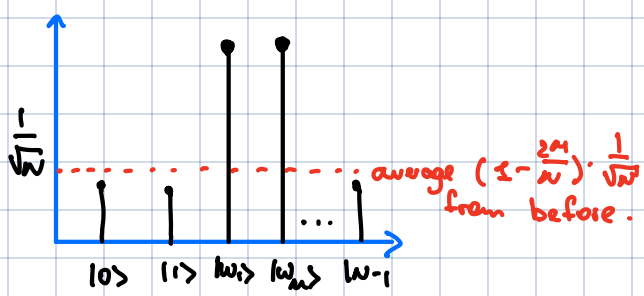
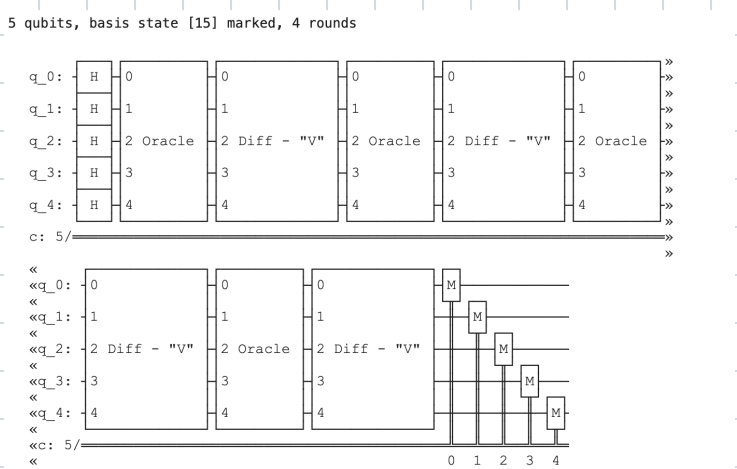
1.)



2.)



3.)



NOW WE WILL CONSIDER HOW TO DO THIS IN QISKIT.